MD Ziauddin

Senior Software Engineer | AI Engineer | Forward Deployed Engineer

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Executive Summary

Senior Software Engineer and Forward Deployed AI Engineer with 5+ years of experience architecting full-stack platforms, real-time distributed systems, and production-grade Voice AI & GenAI workflows. Proven track record of scaling enterprise Voice AI deployments to 1,000,000+ outbound calls, reducing fintech platform latency by 90%, and delivering mission-critical software for global enterprises including VMware. Adept at bridging deep backend engineering, LLM/RAG orchestration, and customer-facing solution architecture.

Technical Skills

AI, GenAI & Voice Systems: LLMs, RAG, Agentic Workflows, LangGraph, LangChain, Voice AI (STT/TTS Benchmarking, Conversational Latency Mitigation), Automated LLM Evaluation Pipelines, AWS Bedrock, OpenAI APIs, Vector Databases (Qdrant, Weaviate)
Languages, Backend & Architecture: TypeScript, JavaScript, Python, SQL, Node.js, Fastify, Express.js, WebSockets, Event-Driven Architecture, REST APIs, OAuth 2.0, Microservices, System Design
Frontend & Visualization: React.js, Next.js, Tailwind CSS, AntV G6 (Graph Visualization), Google Apps Script (React GAS Framework), Figma-to-Code UI Systems
Databases, Cloud & Enterprise Integrations: PostgreSQL, Redis, RabbitMQ, Supabase, AWS, GCP, Azure, Docker, Kubernetes, Turborepo, Enterprise CRMs & Telephony (Odoo, LeadSquared, Twilio, Exotel, Ozonetel)

Professional Experience

OmniDimension Kolkata, West Bengal, India
Forward Deployed Engineer (FDE) *Feb 2026 – Oct 2026*

- **Accelerated Promotion & Ownership:** Promoted from Software Engineer within 3 months to spearhead enterprise Voice AI deployments, owning end-to-end solution architecture from design through production launch and optimization.
- **High-Scale Production Voice AI:** Architected and deployed autonomous Voice AI workflows for enterprise healthcare campaigns, reliably processing **1,000,000+ outbound calls** and driving multi-account expansion.
- **LLM, STT & TTS Benchmarking:** Profiled and benchmarked diverse combinations of LLMs, Speech-to-Text (STT), and Text-to-Speech (TTS) models to establish optimal latency-accuracy configurations, accelerating domain-specific customer onboarding.
- **Enterprise CRM & API Integrations:** Engineered bidirectional integrations between Voice AI agents, scheduling engines, and enterprise CRMs (**Odoo, LeadSquared**), generating **50+ qualified patient consultations** in month one.
- **Production Triage & Launch Leadership:** Served as primary engineering liaison for enterprise e-commerce Voice AI rollouts, leading live production triage, root-cause validation, and cross-team incident resolution.

Software Engineer – GenAI *Nov 2025 – Feb 2026*

- **Conversational Latency Mitigation:** Engineered a dynamic contextual filler system to mask LLM inference and backend network latency, significantly reducing perceived response wait times during live voice calls.
- **Automated AI Evaluation Pipeline:** Built an automated LLM-based QA and compliance pipeline to audit call recordings against operational SOPs and proprietary scoring matrices at scale.
- **Event-Driven Alerting & Analytics:** Designed a real-time, rule-based notification framework (broadcast and user-specific alerts) and automated campaign analytics reporting infrastructure for enterprise stakeholders.

GROWWTH PARTNERS Kolkata, West Bengal, India
Senior Software Engineer *Jul 2024 – Nov 2025*

- **Platform Modernization & Performance:** Architected the core overhaul of an AI-powered fintech platform, **reducing application latency by 90%** and **accelerating release velocity by 40%** through scalable system design.
- **Real-Time AI Assistant Architecture:** Designed and implemented a bidirectional **WebSocket** communication layer, transforming a static web application into a low-latency, interactive AI assistant experience.
- **End-to-End System Ownership:** Owned technical architecture, code reviews, and multi-release delivery for flagship AI Assistant and Referral Engine initiatives across product and engineering teams.
- **Developer Experience & Mentorship:** Created a standardized React Google Apps Script (GAS) framework to unify frontend development, while mentoring **4 junior engineers** in system design and code quality.

Copods Pune, Maharashtra, India
Software Engineer *Aug 2021 – Apr 2024*

- **VMware Enterprise Licensing Platform (Mar 2022 – Aug 2023):** Migrated legacy authentication infrastructure to **OAuth 2.0**, securing core enterprise APIs and streamlining downstream developer integrations.
- **Enterprise UI Modernization (VMware):** Served as primary UI engineer for VMware's global branding modernization, re-architecting customer-facing licensing and subscription workflows in **React** for a global enterprise user base.
- **Ariksa Enterprise IT Visualization (Aug 2021 – Mar 2022):** Engineered high-performance infrastructure monitoring dashboards and interactive VM dependency relationship graphs using **AntV G6** and scalable backend APIs.

Education & Certifications

Future Institute of Engineering and Management Kolkata, West Bengal, India
Bachelor of Technology (B.Tech) in Information Technology *Jul 2017 – Jul 2021*

- **Certifications & Continuous Learning:** Become an AI Engineer (*LinkedIn*), Google & Kaggle 5-Day GenAI Intensive, Google Cloud Machine Learning Engineer Learning Path.